

MoBI-View: An Open-Source Realtime Biosignal Visualization Tool for LSL

Cross-Platform Monitoring and Extensible Visualization of Synchronized Multimodal LSL Streams



Iktae Kim, Adam Santorelli, John Vito d’Antonio-Bertagnolli, Stan Colcombe, Alexandre Franco, Michelle Freund, Michael Milham
Child Mind Institute, New York

Motivation

Mobile Brain/Body Imaging (MoBI) combines EEG, eye tracking, and physiological sensing during naturalistic tasks.

- Can we easily monitor that every data stream is present, correctly configured, and producing sensible data intra-session?

Problems can be detected too late

Visualization is often fragmented

Native GUI coupling adds maintenance

Design Objective: provide one open-source browser interface for monitoring synchronized numeric LSL streams in real time, while keeping acquisition and visualization independently maintainable.

Challenges / Methods

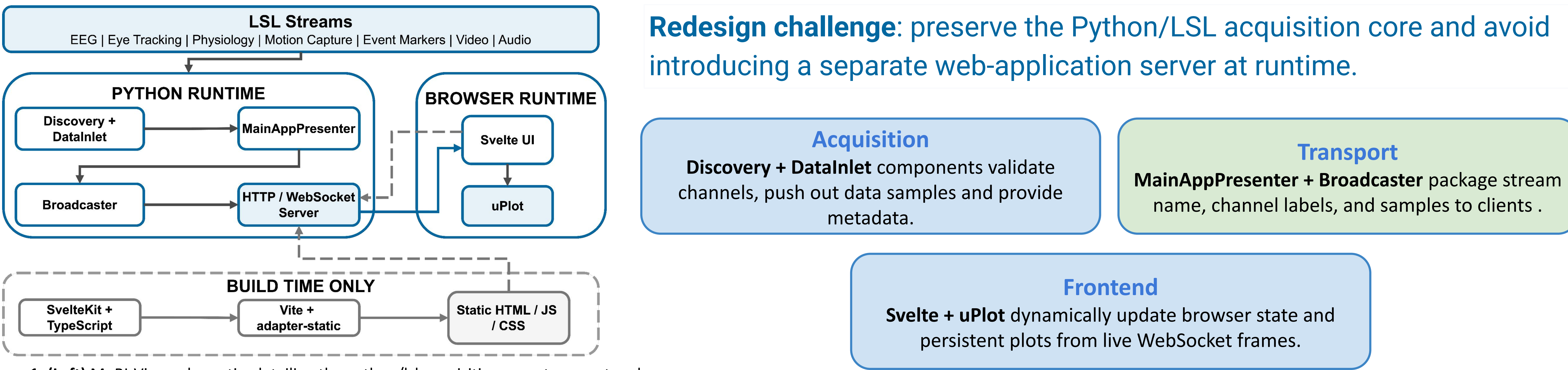


Figure 1. (Left) MoBI-View schematic, detailing the python/lsL acquisition core, transport and frontend. (Right) Brief description of these three phases.

Results

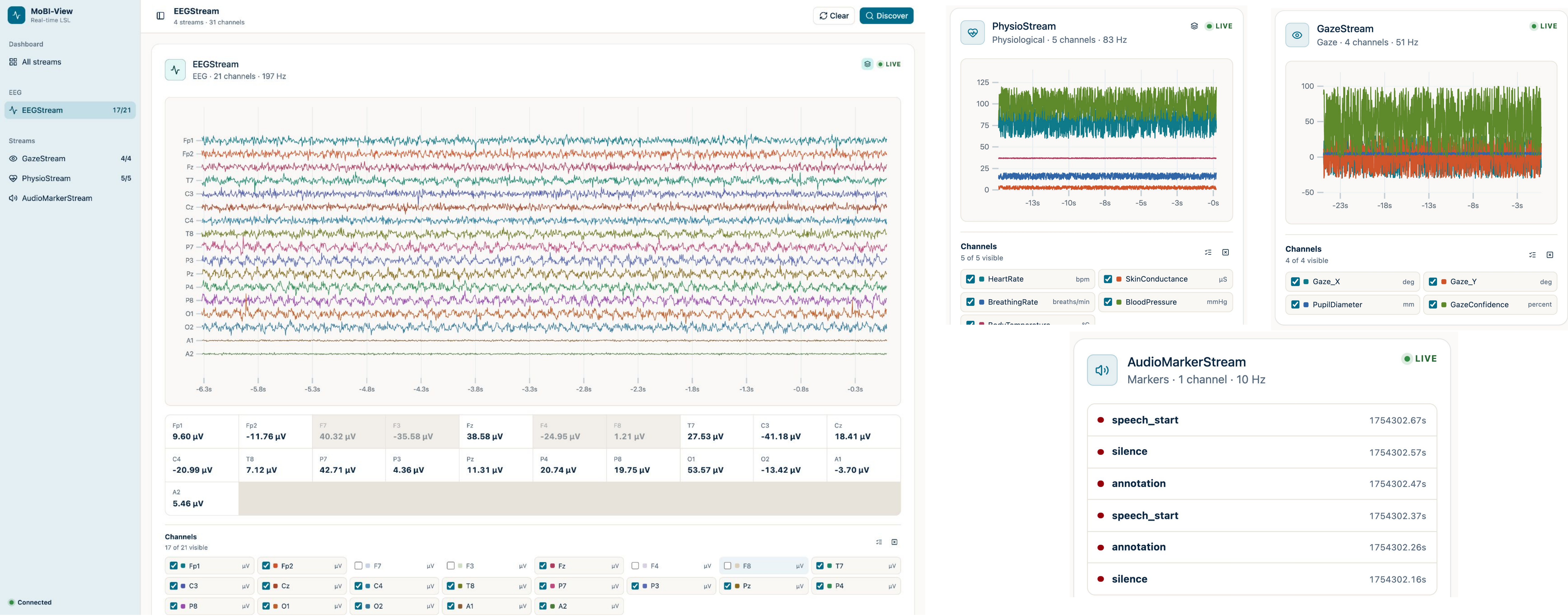


Figure 2. (Left) Browser-based MoBI-View interface showing discovered LSL streams, interactive stream/channel controls, and real-time uPlot visualization of incoming multimodal signals, with focus on EEG plot. (Right) Configurable additional widgets for eye-tracking, physio signals, and Audio-Visual streams.

Take home message

MoBI-View provides a unified, open-source platform for monitoring synchronized multimodal LSL data in real time, helping researchers inspect signals across modalities, identify acquisition problems during a session, and verify experimental protocols.

SCAN ME !



Contact information:
Iktae Kim - iktae.kim@childmind.org
John Vito - john.vito@childmind.org
Michael Milham - michael.milham@childmind.org

References:
[1] Gramann, K. (2019). Mobile Brain/Body Imaging (MoBI).
[2] Kothe, 2014 – Lab Streaming Layer (LSL)